Machine Learning (69152)

Bayesian Optimization for policy search

Rubén Martínez Cantín
Dpto. Informática e Ingeniería de Sistemas.

Content

- **Bayesian learning**
- **Bayesian regression**
  - Bayesian linear regression
  - Gaussian parametric regression
- **Gaussian processes**
  - GPs for regression
  - Kernels
  - GPs beyond regression
- **Bayesian inference**
  - Monte Carlo
  - Markov Chain Monte Carlo

(Decision making and RL Break)

- **Bayesian decision making**
  - Active learning
  - Bayesian optimization

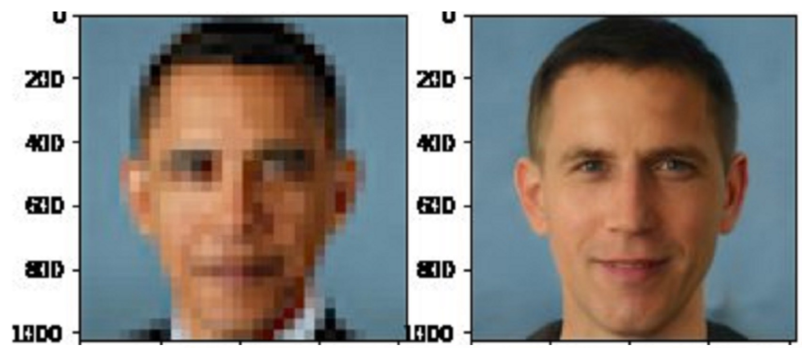
The fallacy of Big Data

- In 2016, Tesla had **12 billion kilometers** of data.
- Enough for self-driving?

The problems of Big Data

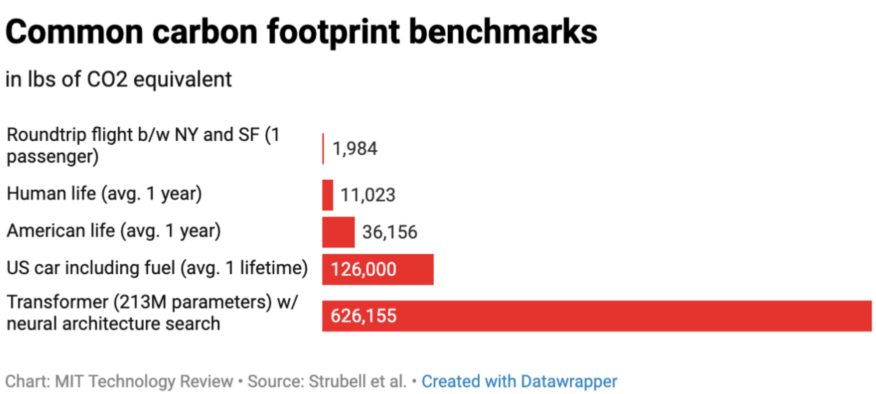
- Personal data collection is out of control.
  - Check the permissions of your phone apps or the cookies of a website.
  - Some data collection is immoral. Data is biased. Watch out for human “cleaning”.

Three black/white teenagers



The problems of Big Data

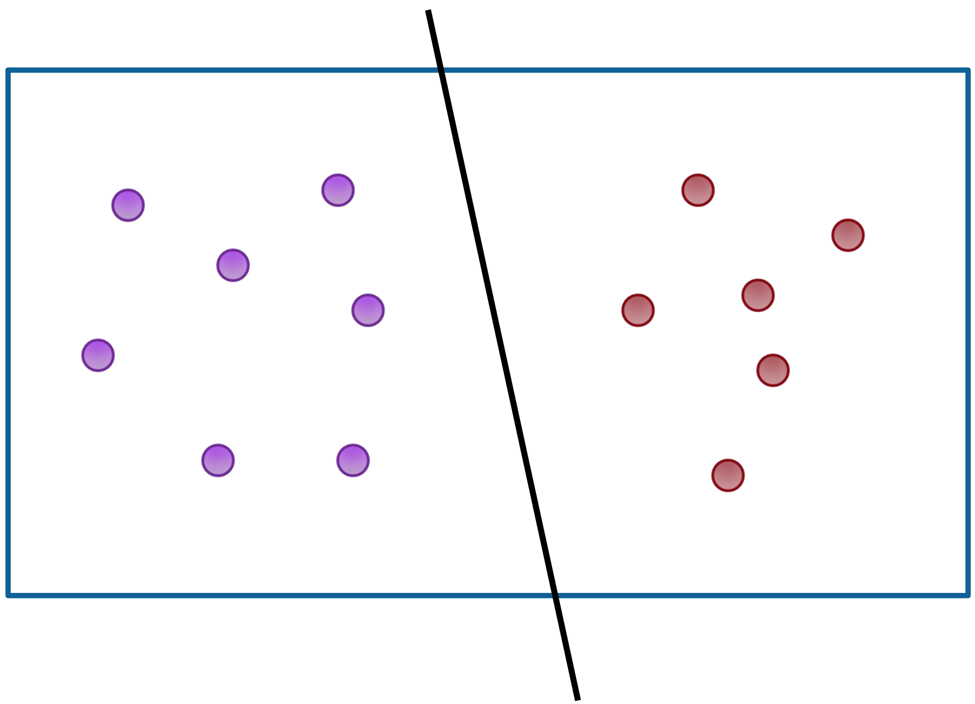
- Big data management and learning is costly.
  - Exploiting “ghost workers”.
  - High energy consumption.



- We need new paradigm of Small Data.

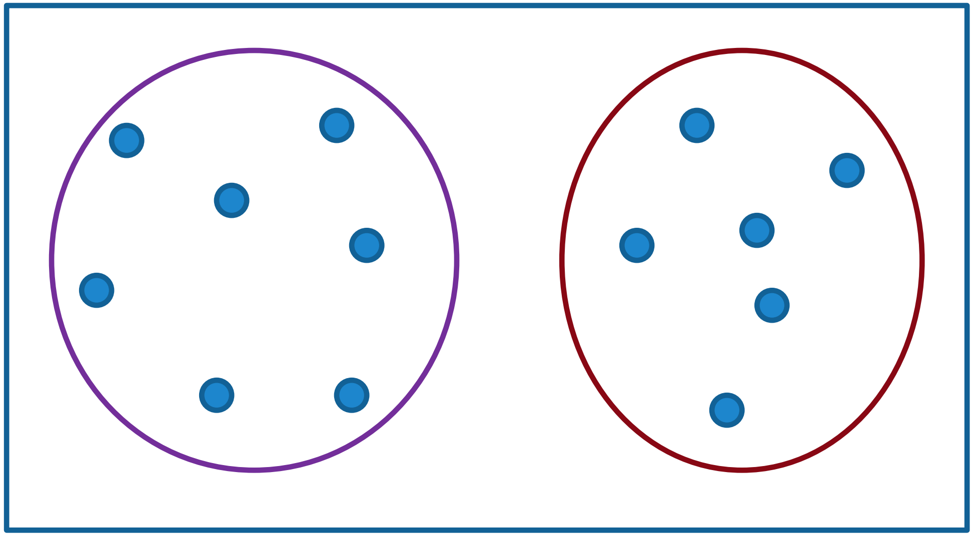
Learning:

- Supervised learning:
  - Based on labels.



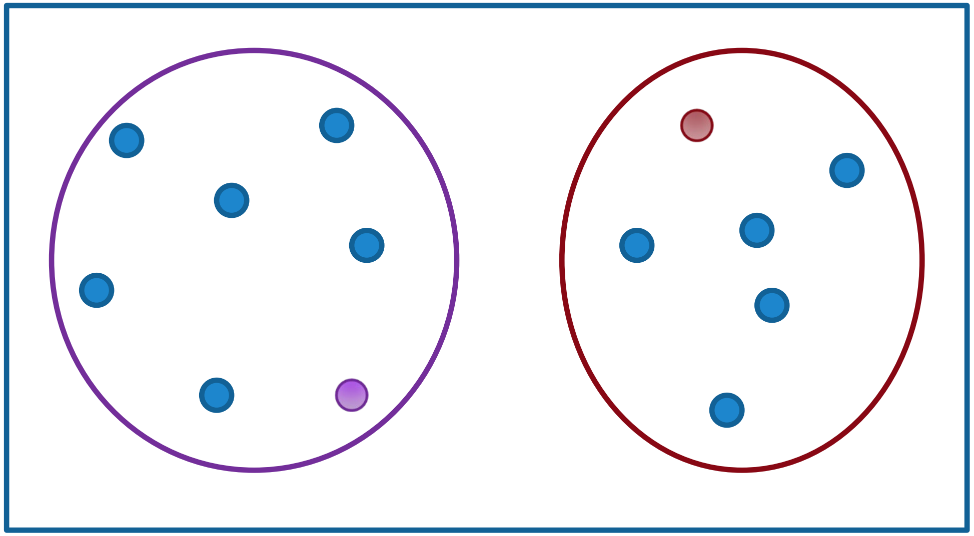
Learning:

- Unsupervised learning:
  - Based on structure.



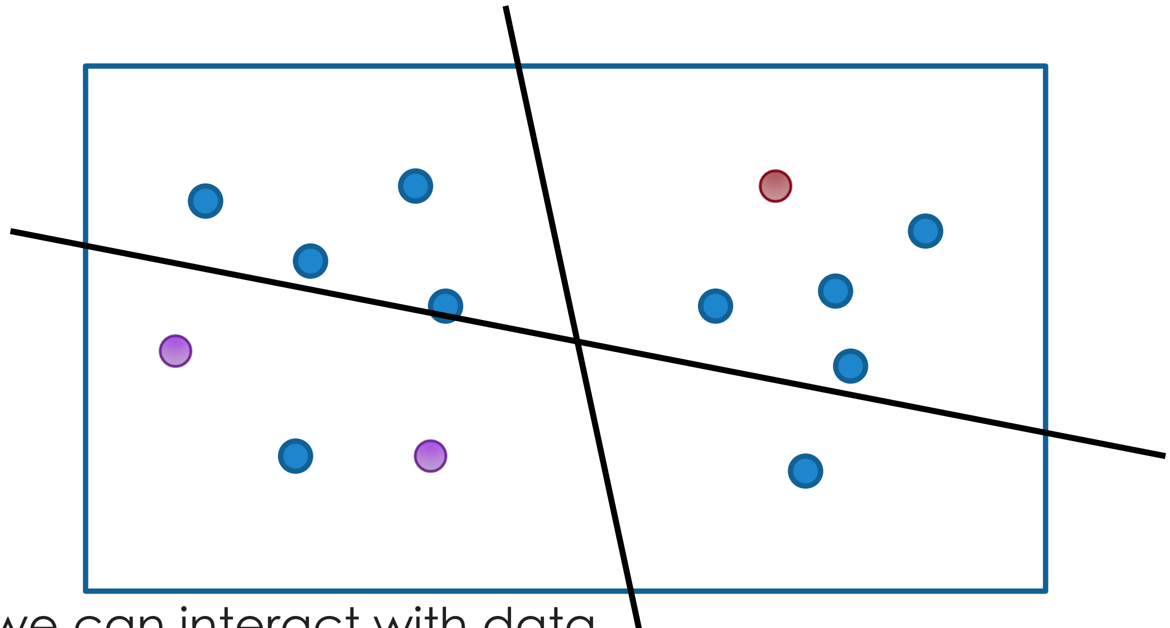
Learning:

- Semi-supervised learning:
  - Based on labels and structure.



Learning:

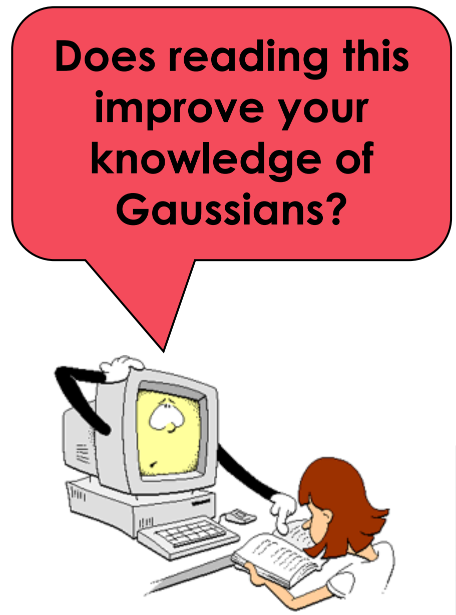
- Active learning:
  - Based on labels, structure and learning algorithm



  - And we can interact with data

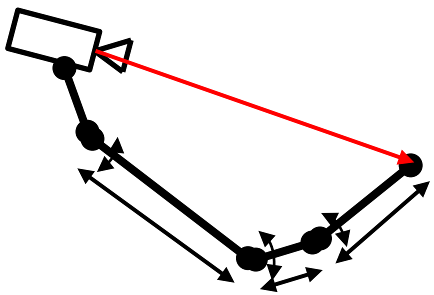
Active Learning: Asking the Right Questions

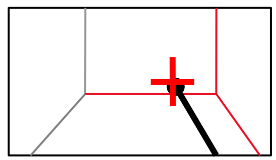
- In **active learning**, the machine can query the environment. That is, it can ask questions.
- Decision theory leads to optimal strategies for choosing when and what questions to ask in order to gather the best possible data. **Good data is often better than a lot of data.**
- Humans also query the environment continuously…

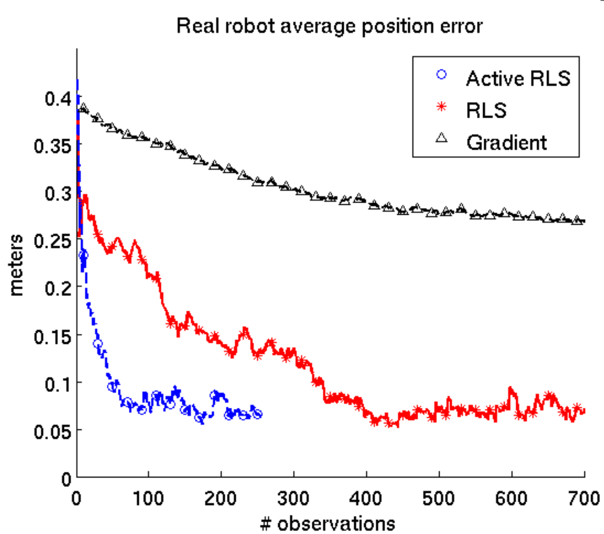


Body Schema Learning

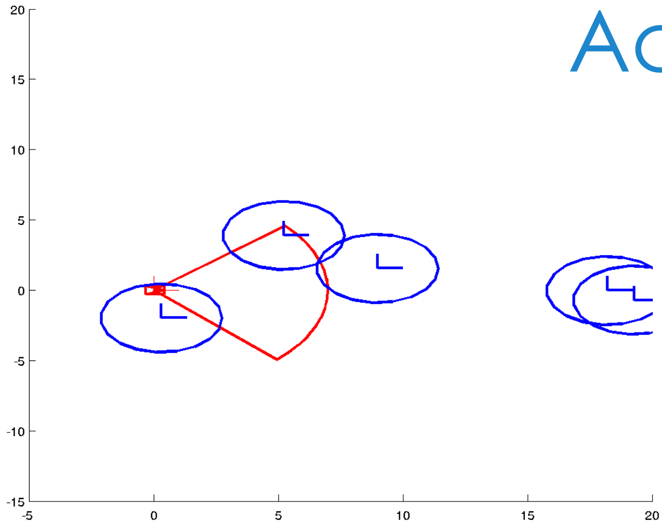
Given propioception (e.g.: motion commands) and extereoception (e.g.: visual cues), estimate the geometric parameters that define our own body.

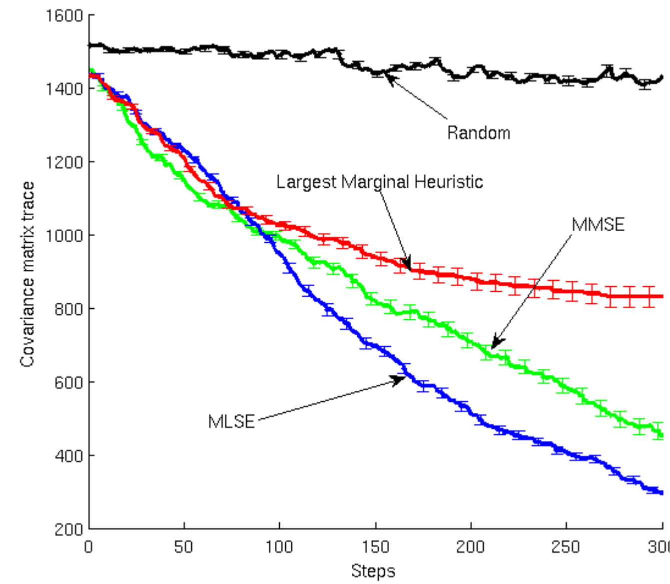




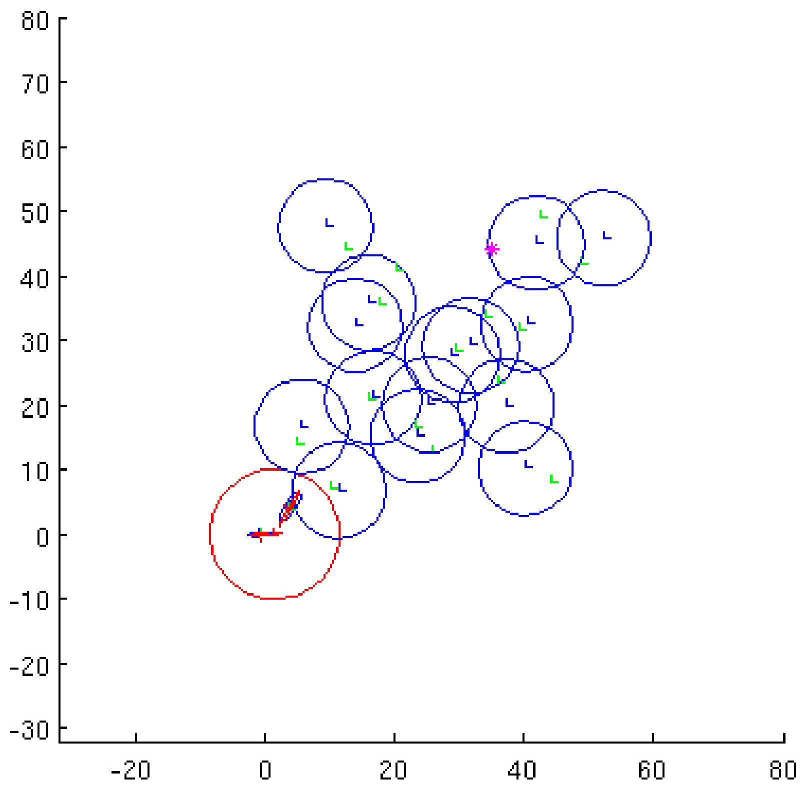


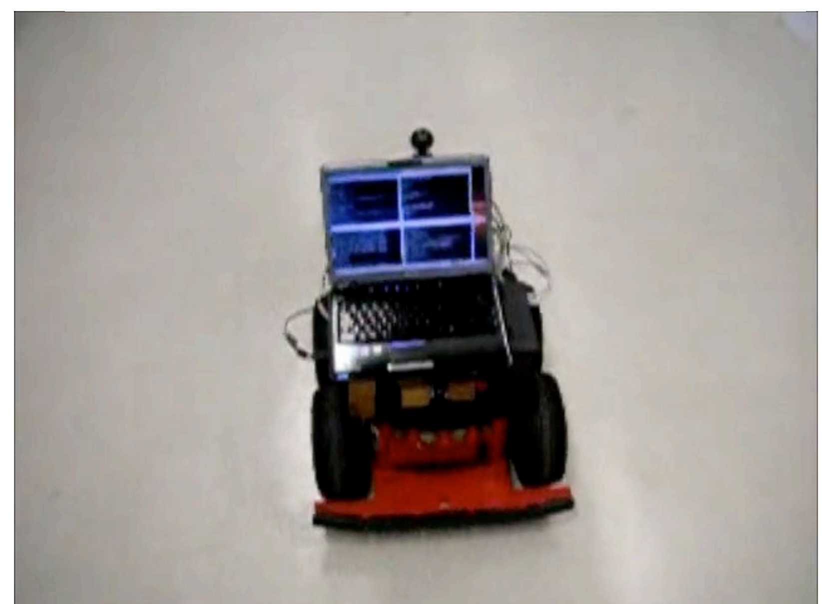
Active SLAM

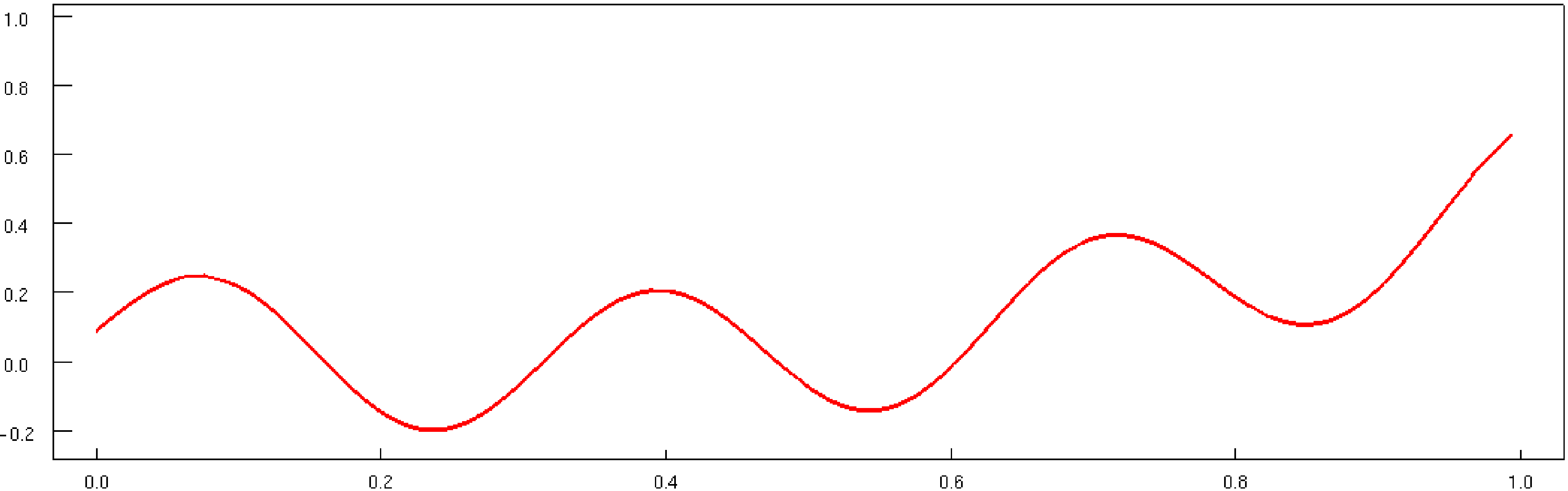




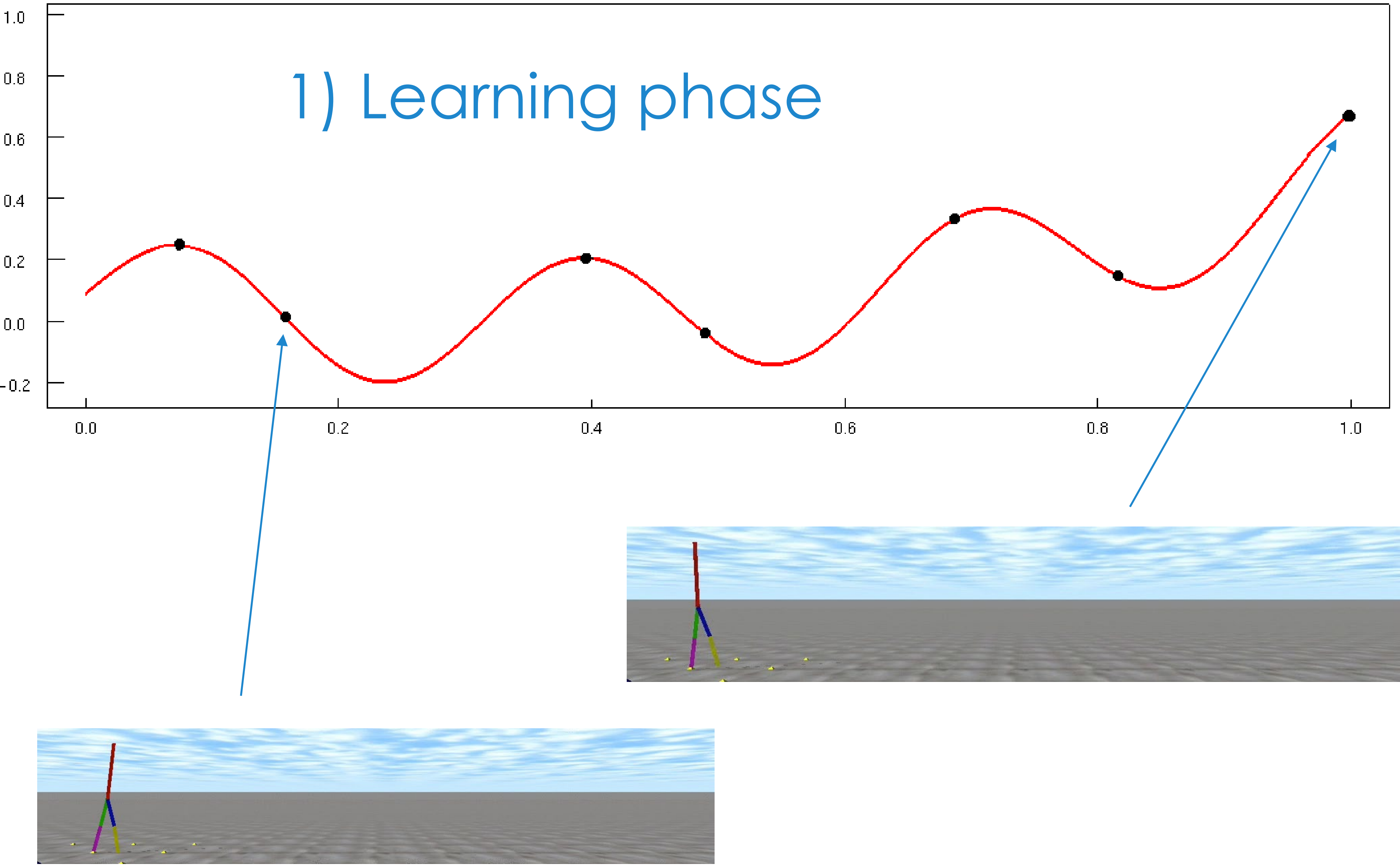
Covariance matrix trace
Steps
Random
Largest Marginal Heuristic
MMSE
MLSE



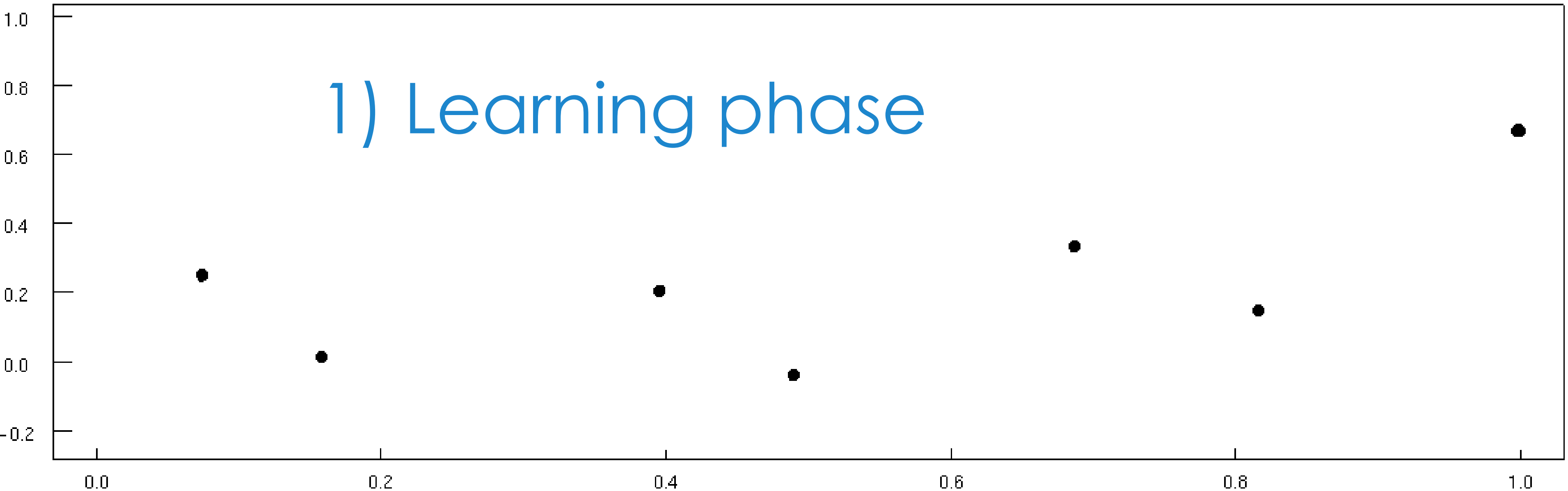


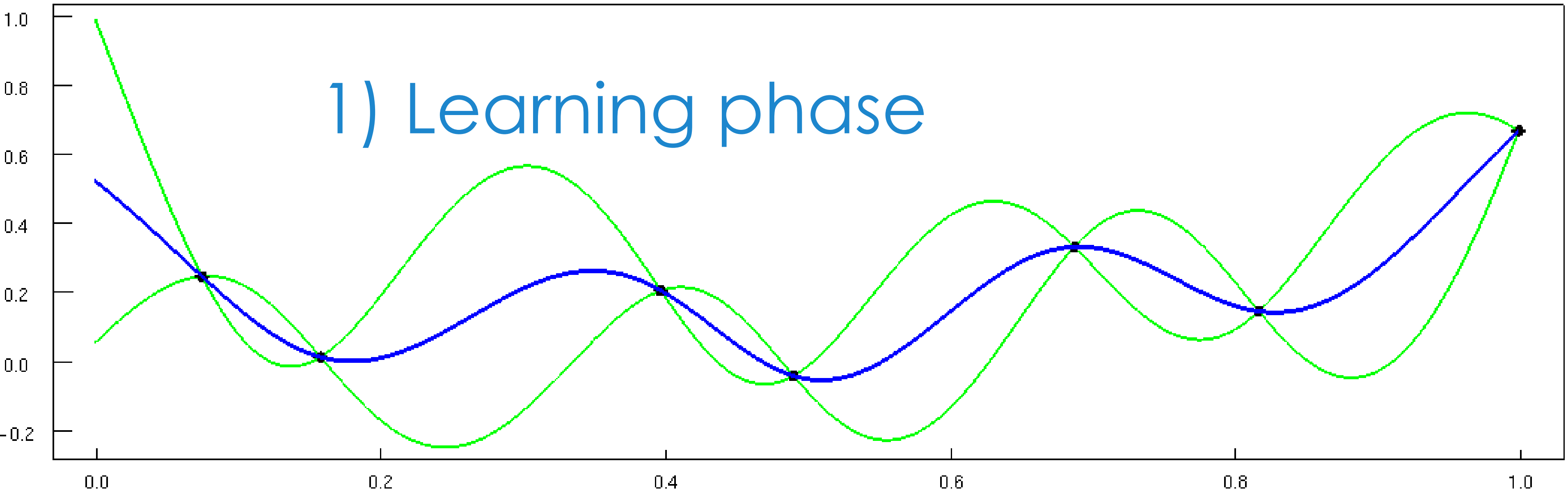


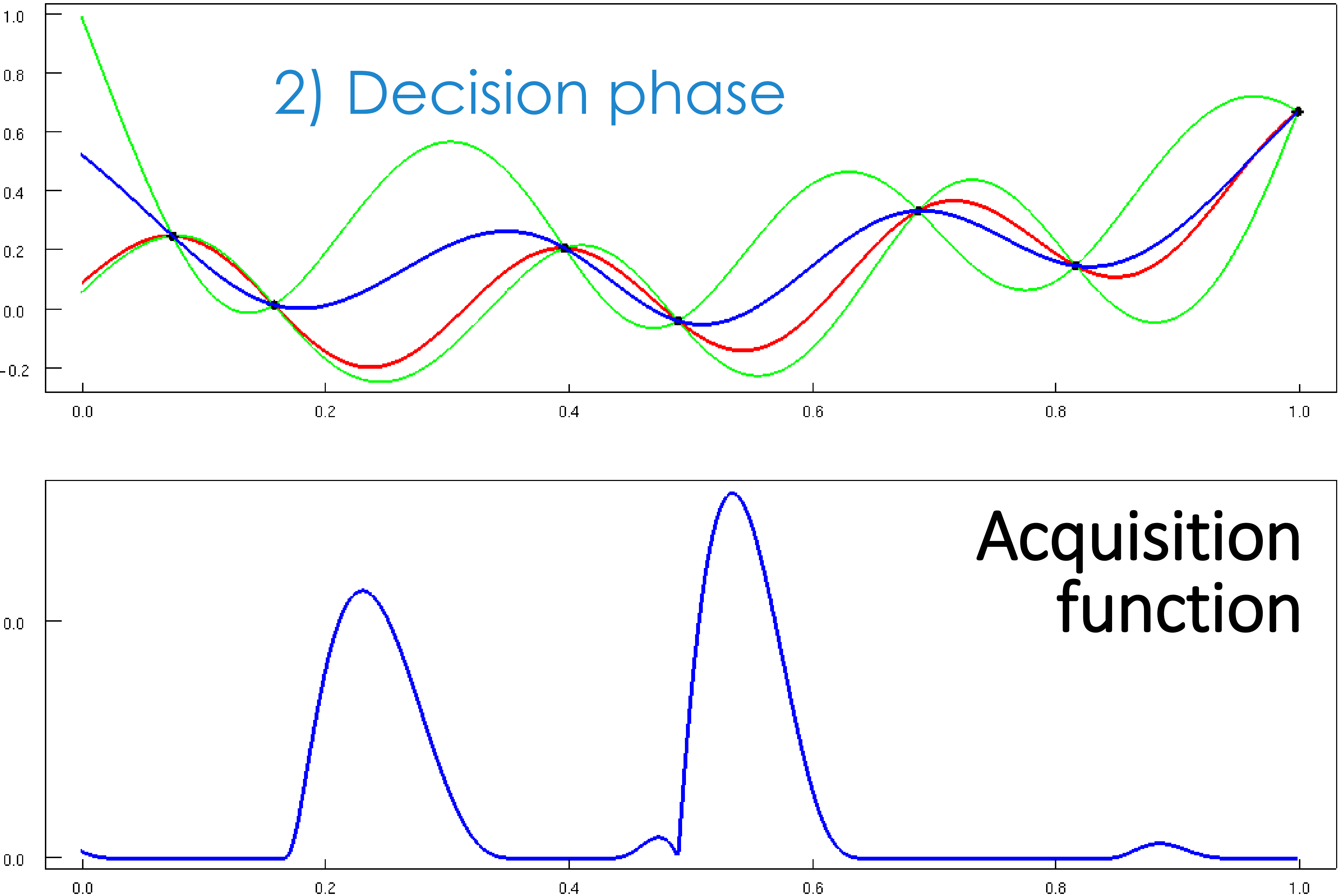
Active policy search: Bayesian optimization

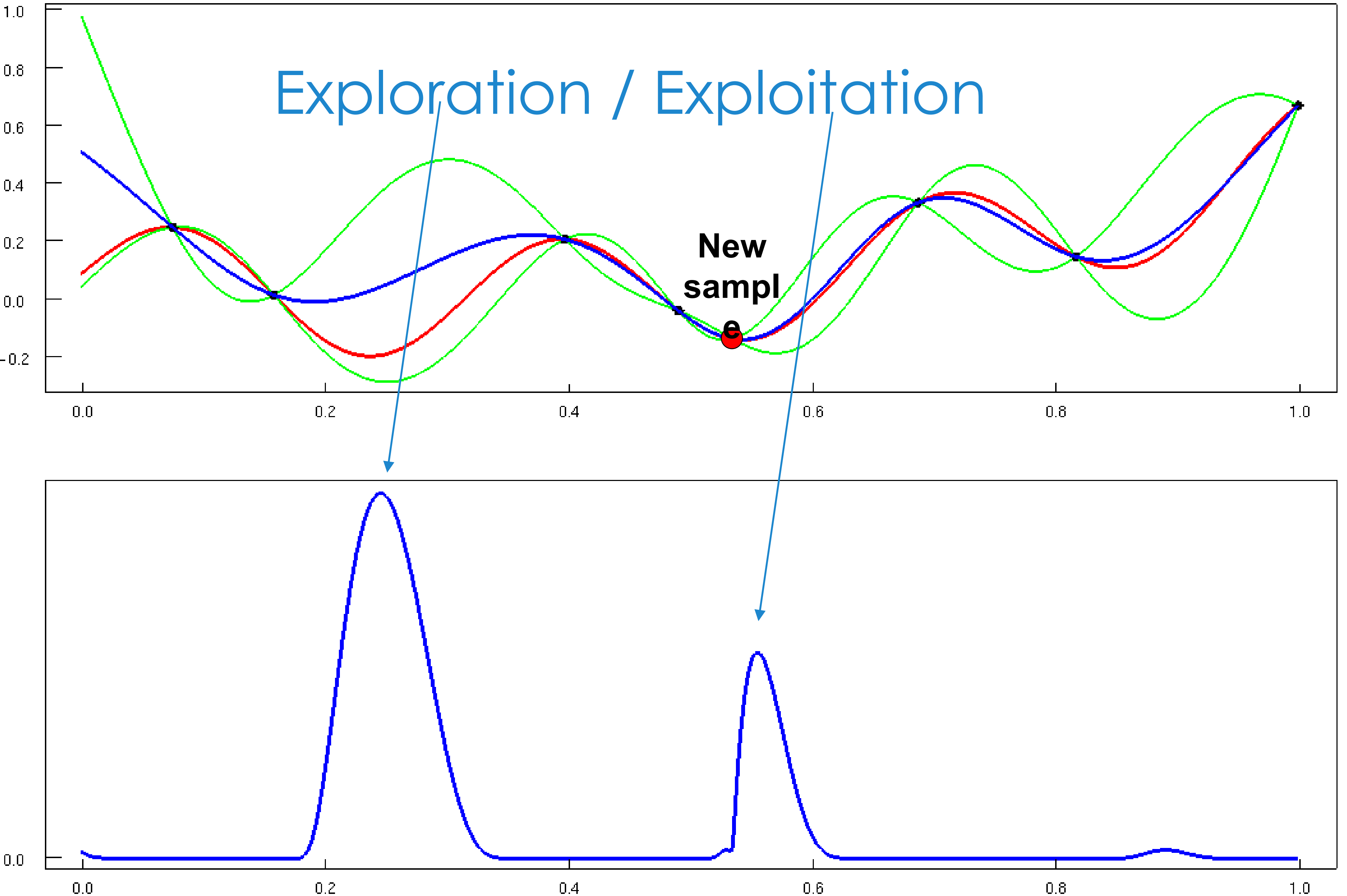

1) Learning phase
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
0.0
0.2
0.4
0.6
0.8
1.0

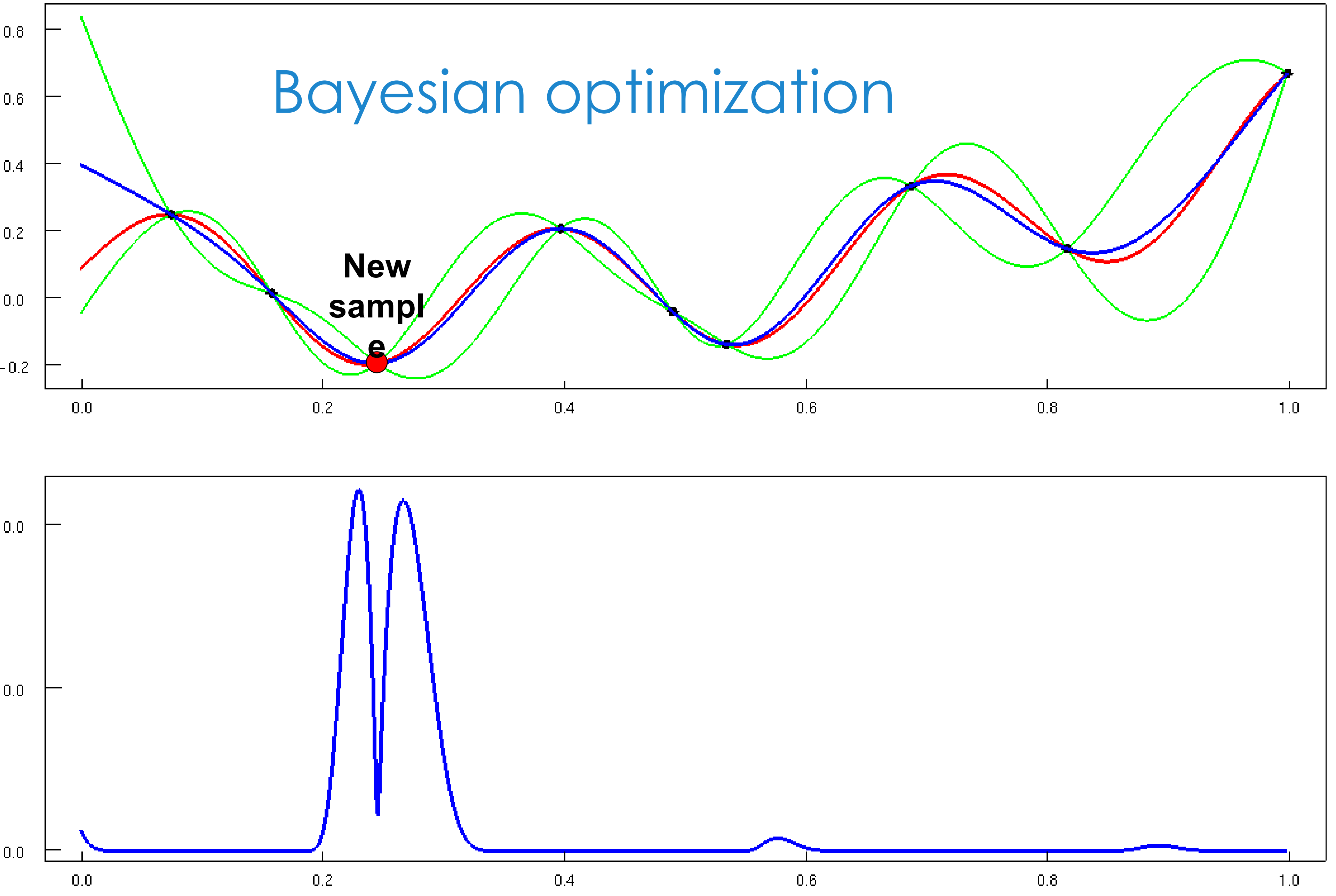
1) Learning phase




1) Learning phase
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
0.0
0.2
0.4
0.6
0.8
1.0


2) Decision phase
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
0.0
0.2
0.4
0.6
0.8
1.0
Acquisition function
0.0
0.0
0.0
0.2
0.4
0.6
0.8
1.0


Exploration / Exploitation
New sample
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.0
0.2
0.4
0.6
0.8
1.0



Bayesian optimization
New sampl e
0.8
0.6
0.4
0.2
0.0
-0.2
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.0
0.0
0.0
0.2
0.4
0.6
0.8
1.0

Bayesian optimization as expected utility

- Utility → e.g.: improvement function

$$I(x) = \max\{0, f(x) - f(x_{best})\}$$

- Distribution → e.g.: Gaussian process

$$P(f) \sim \mathcal{GP}\left(0, \mathbf{K}(\theta) + \sigma_n^2 \mathbf{I}\right)$$

- Expected utility → e.g.: Expected improvement

$$\mathbb{E}[I(x)] = \underbrace{(\mu(x) - f(x_{best}))}_{exploit} \Phi(d) + \underbrace{\sigma^2(x)}_{explore} \phi(d)$$

$$d = \frac{x - \mu(x)}{\sigma(x)}$$

Universidad Zaragoza 1542

How does it work?

Input: prior $P(f)$, criterion $\mathcal{C}(\cdot)$, budget N **Output:** $\mathbf{x}^*$

Build a set of points $\mathbf{X} = \mathbf{x}_1 \ldots \mathbf{x}_l$ and its response $\mathbf{y} = y_1 \ldots y_l$ using a preliminar design.

While $i < N$

- Update the distribution with all data available
  $P(f|\mathbf{X}, \mathbf{y}) \propto P(\mathbf{X}, \mathbf{y}|f)P(f)$
- Select the point $\mathbf{x}_i$ which maximizes the criterion:
  $\mathbf{x}_i = \arg\max \mathcal{C}(\mathbf{x}|P(f|\mathbf{X}, \mathbf{y}))$.
- Augment the data with the new point and response $y_i = f(x_i) + \epsilon$:
  $\mathbf{X} \leftarrow \{\mathbf{X} \cup \mathbf{x}_i\} \qquad \mathbf{y} \leftarrow \{\mathbf{y} \cup y_i\} \qquad i \leftarrow i + 1$

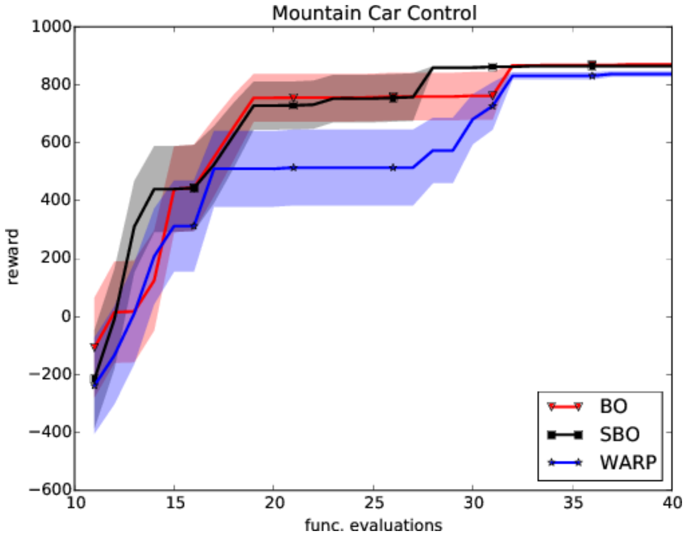
Optimization to solve optimization

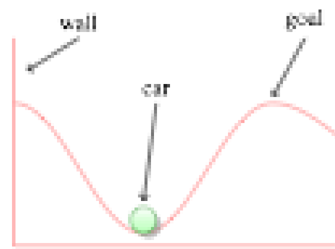
- For each new sample . . .
- . . . select the point $\mathbf{x_i}$ which maximizes the criterion:

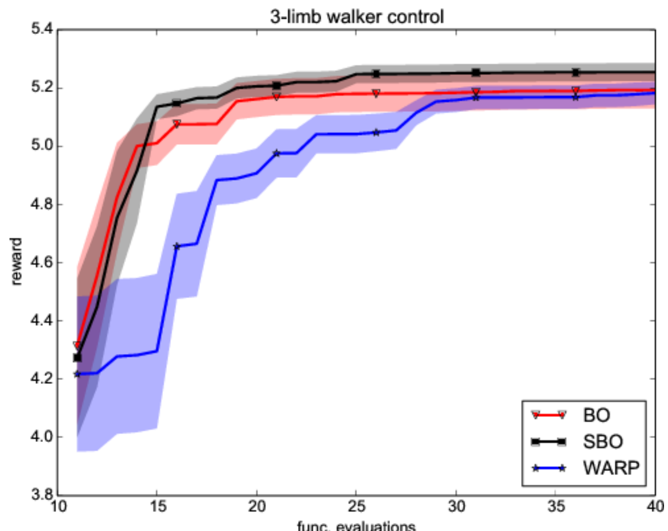
$$\mathbf{x}_i = \arg\max \mathcal{C}(\mathbf{x} | P(f | \mathbf{X}, \mathbf{y}))$$

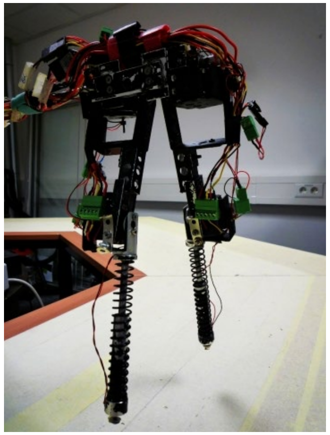
Universidad Zaragoza 1542

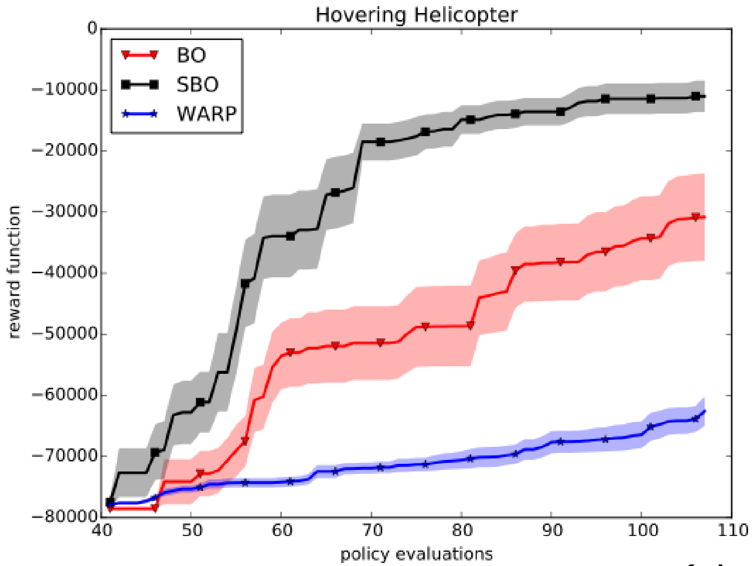
Robot control and policy search



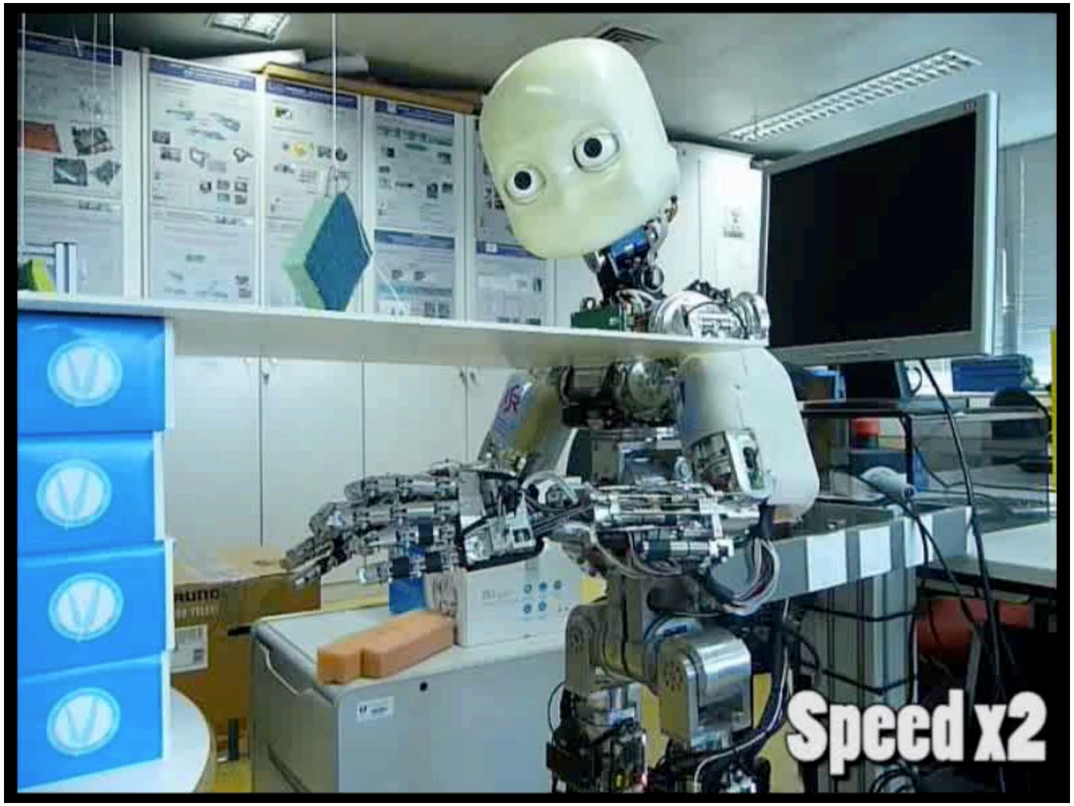


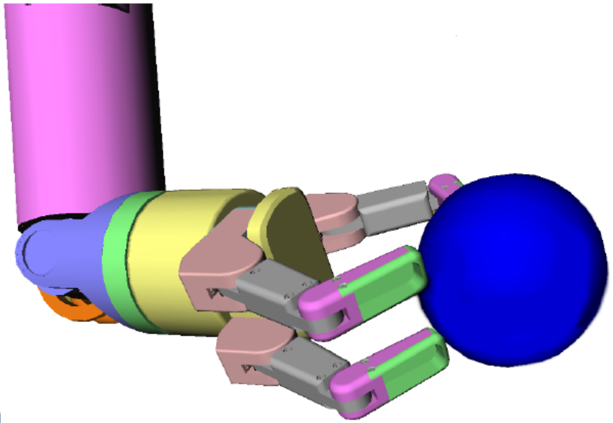


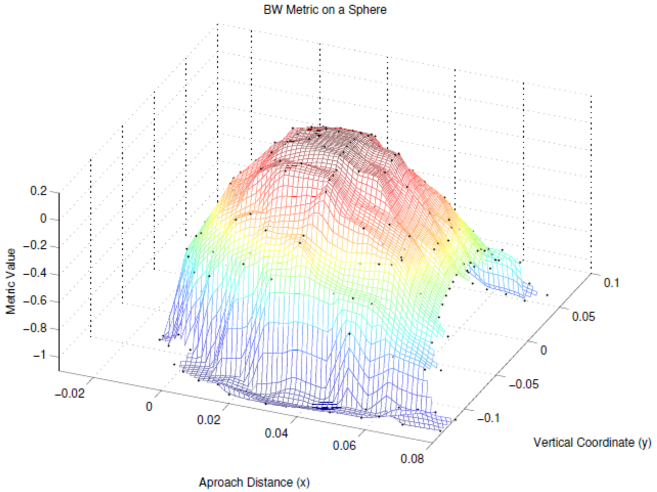




Robot grasping

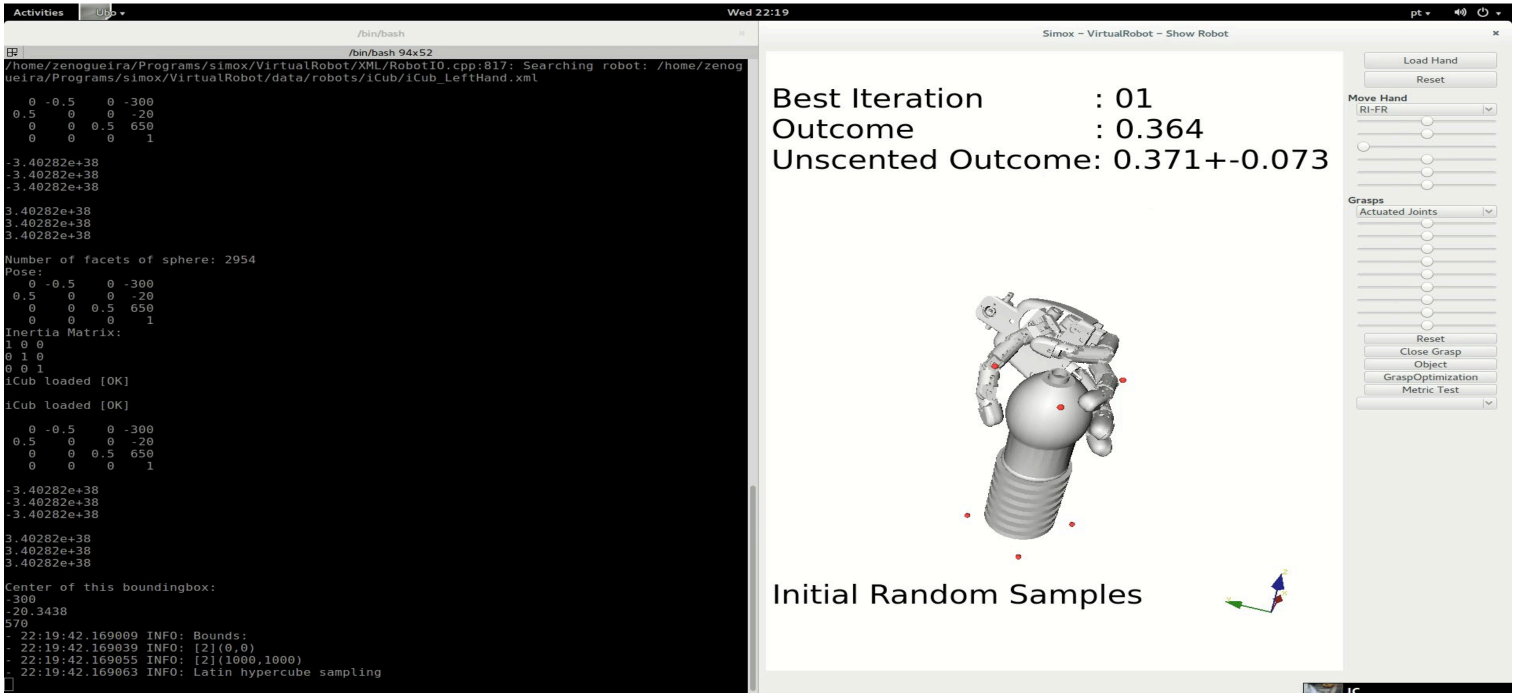


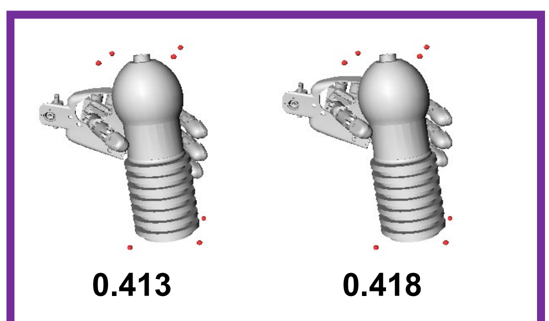


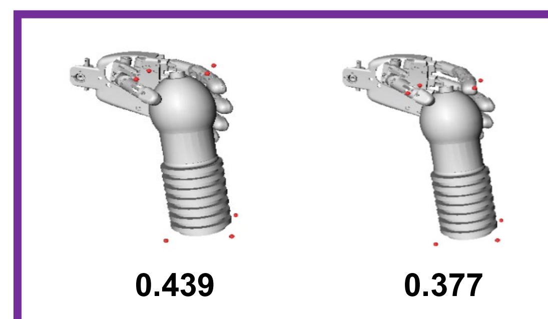


Un Zaragoza

Unscented Bayesian Optimization



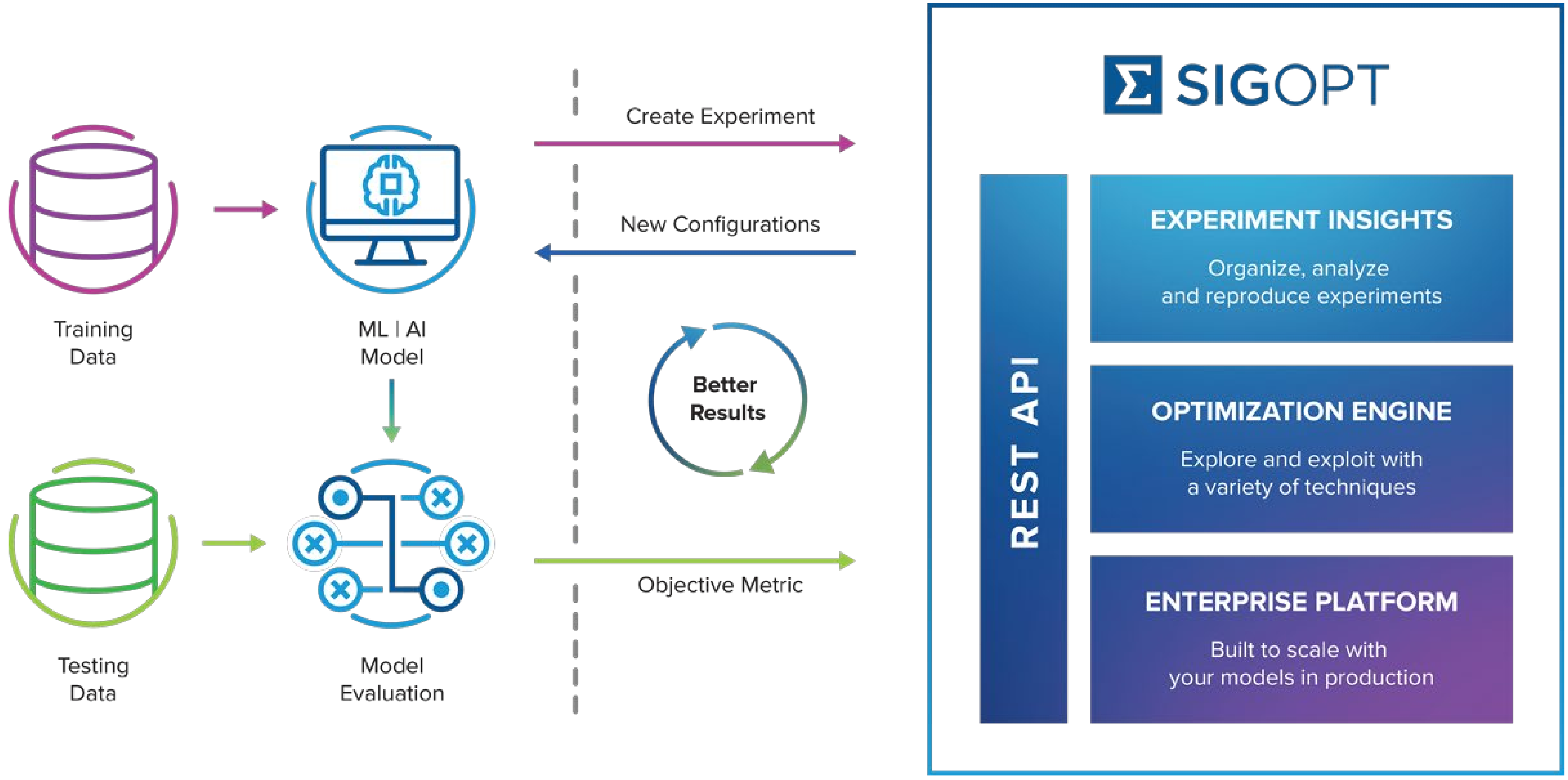




Universidad Zaragoza
1542

Auto-ML

- Hyperparameter tuning / Architecture search



Universidad Zaragoza
1542

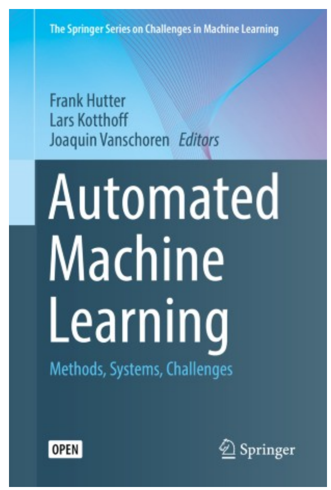
Google Cloud AutoML Vision

Amazon SageMaker

Azure No-Code AutoML

H_2O AutoML

DataRobot



neuralstudio.ai

AutoWeka
An Automated Data Mining Software Based on Weka

OPTUNA

TPOT

Xpanse AI

bigml

AutoKeras

tune

mind FOUNDRY

Universidad Zaragoza
1542

Internal use

NETFLIX

twitter

HUAWEI

facebook

yelp

hotwire

Prudential

BERKELEY LAB

DeepMind

Universidad Zaragoza

1542

Bayesian Optimization in AlphaGo

Yutian Chen, Aja Huang, Ziyu Wang, Ioannis Antonoglou, Julian Schrittwieser, David Silver & Nando de Freitas

DeepMind, London, UK

yutianc@google.com

Abstract

During the development of AlphaGo, its many hyper-parameters were tuned with Bayesian optimization multiple times. This automatic tuning process resulted in substantial improvements in playing strength. For example, prior to the match with Lee Sedol, we tuned the latest AlphaGo agent and this improved its win-rate from 50% to 66.5% in self-play games. This tuned version was deployed in the final match. Of course, since we tuned AlphaGo many times during its development cycle, the compounded contribution was even higher than this percentage. It is our hope that this brief case study will be of interest to Go fans, and also provide Bayesian optimization practitioners with some insights and inspiration.


FLASHBACK

Flashback

- Research project for robotics.

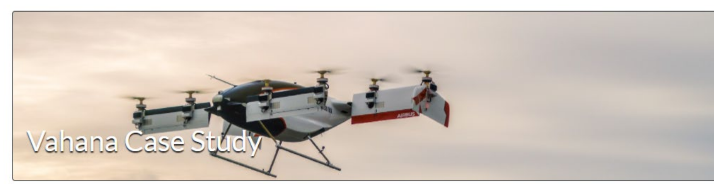
bayesopt Public

BayesOpt: A toolbox for bayesian optimization, experimental design and stochastic bandits.

C++ ☆ 305 ⑂ 78

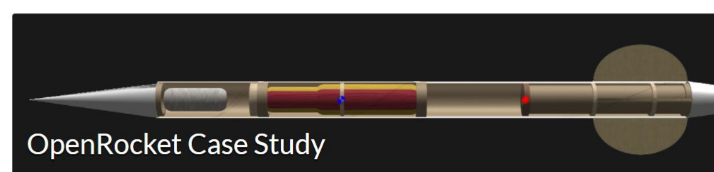
NEXT LIMIT

MetaMorph

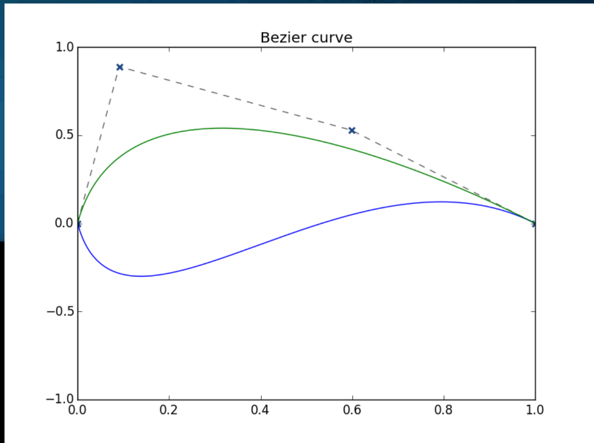


SIGOPT

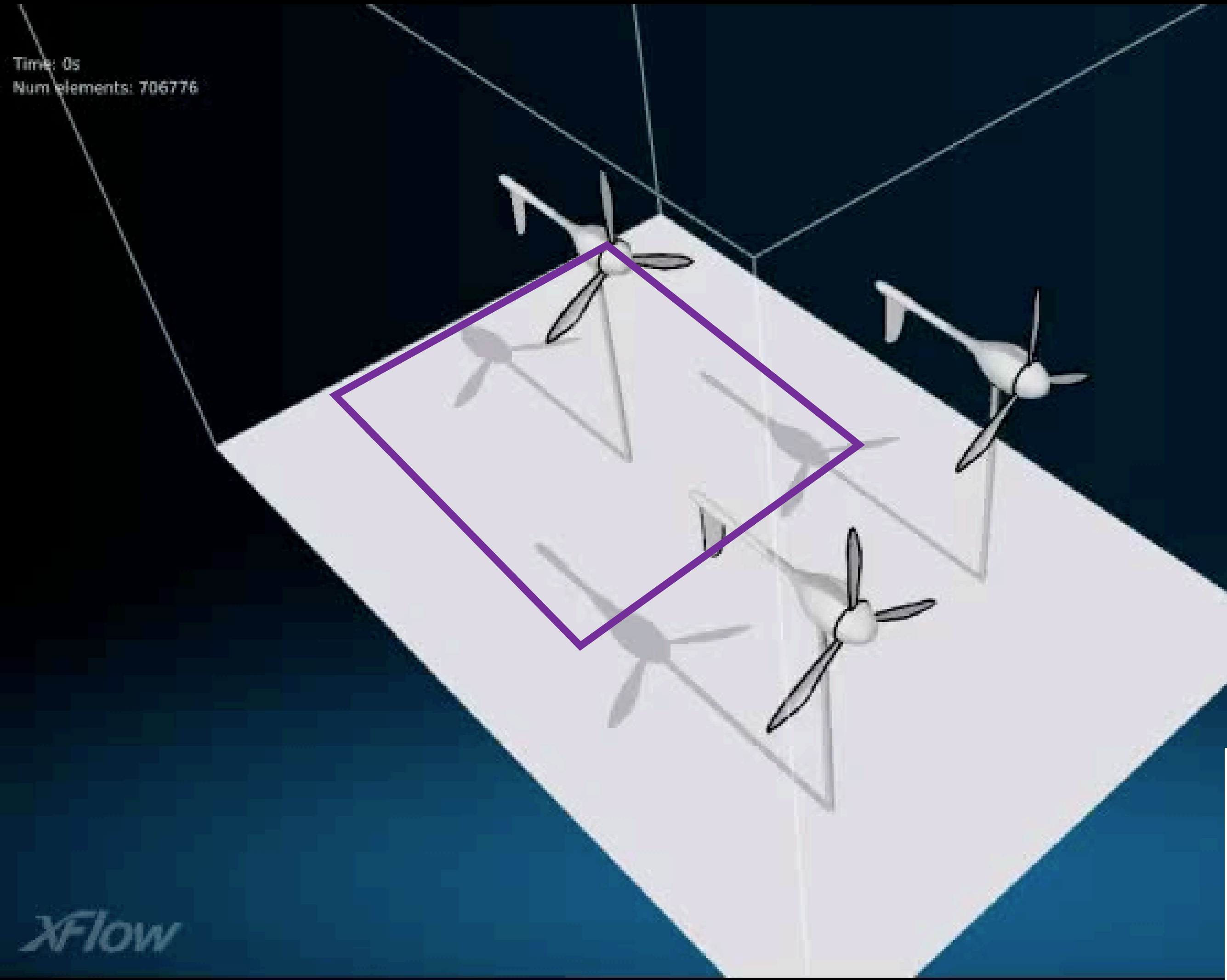
NASA

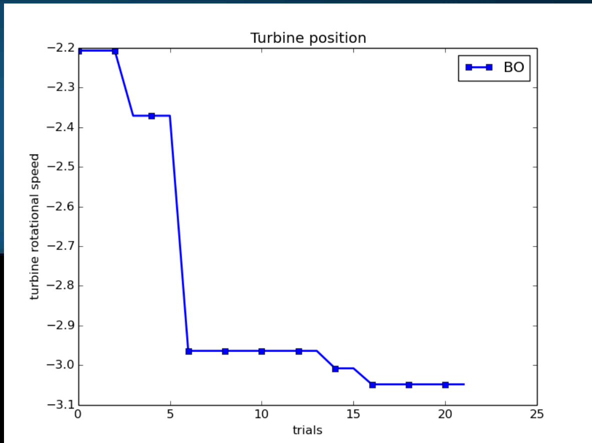


Optimal wing profile design



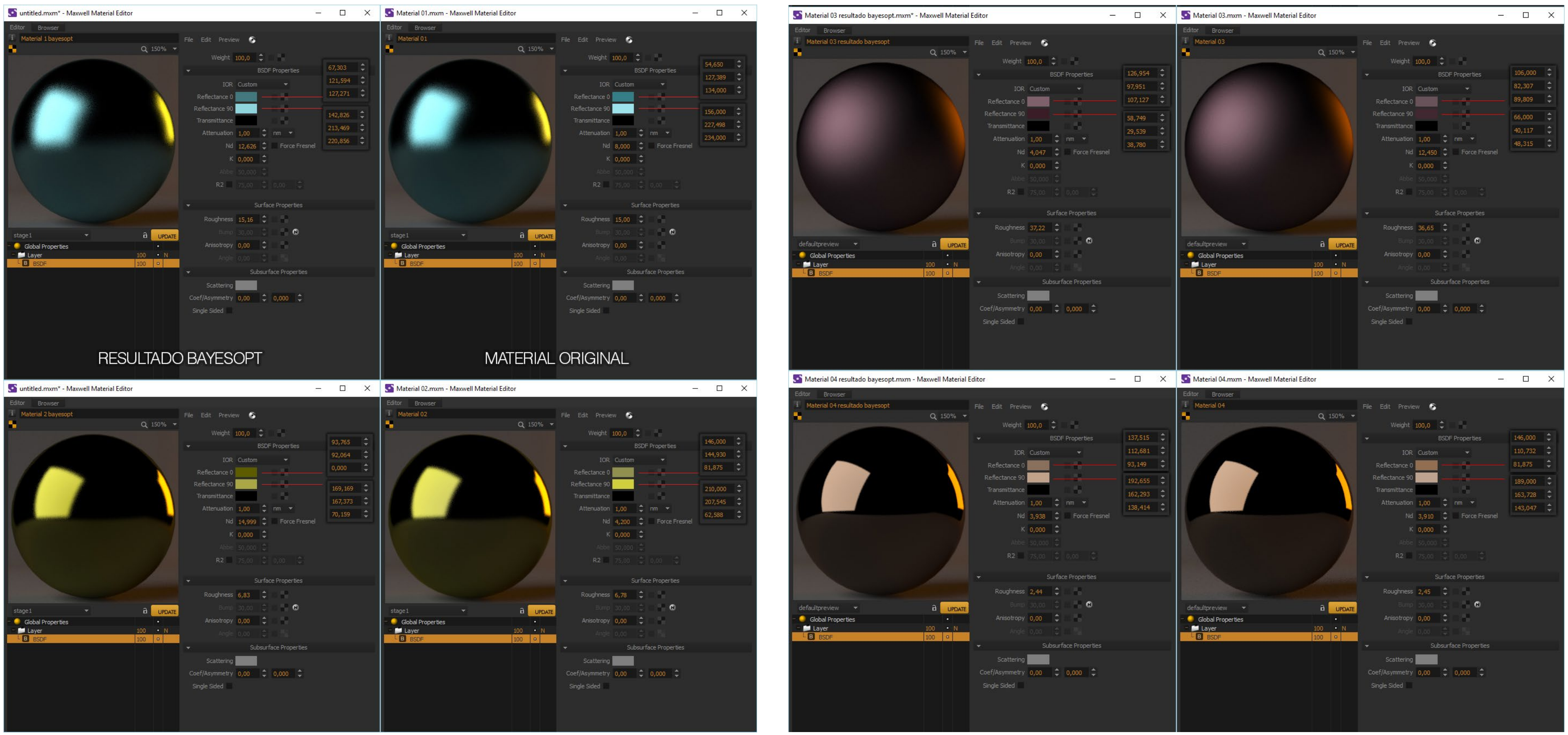
Optimal wind turbine placement



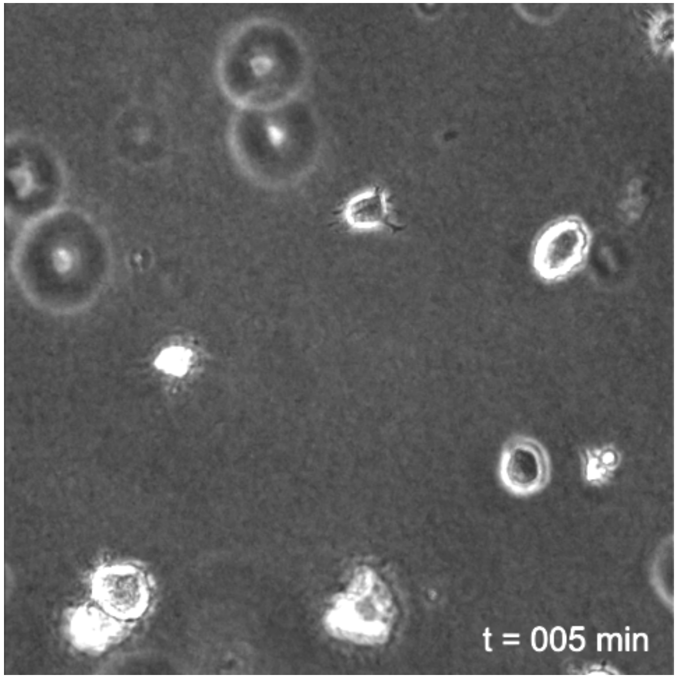


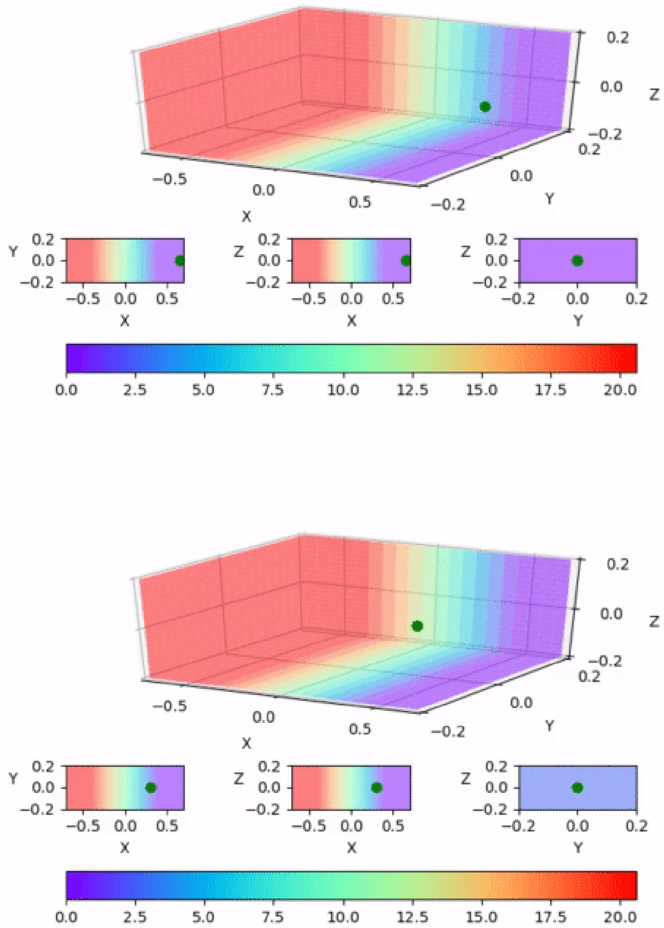
Virtual material matching

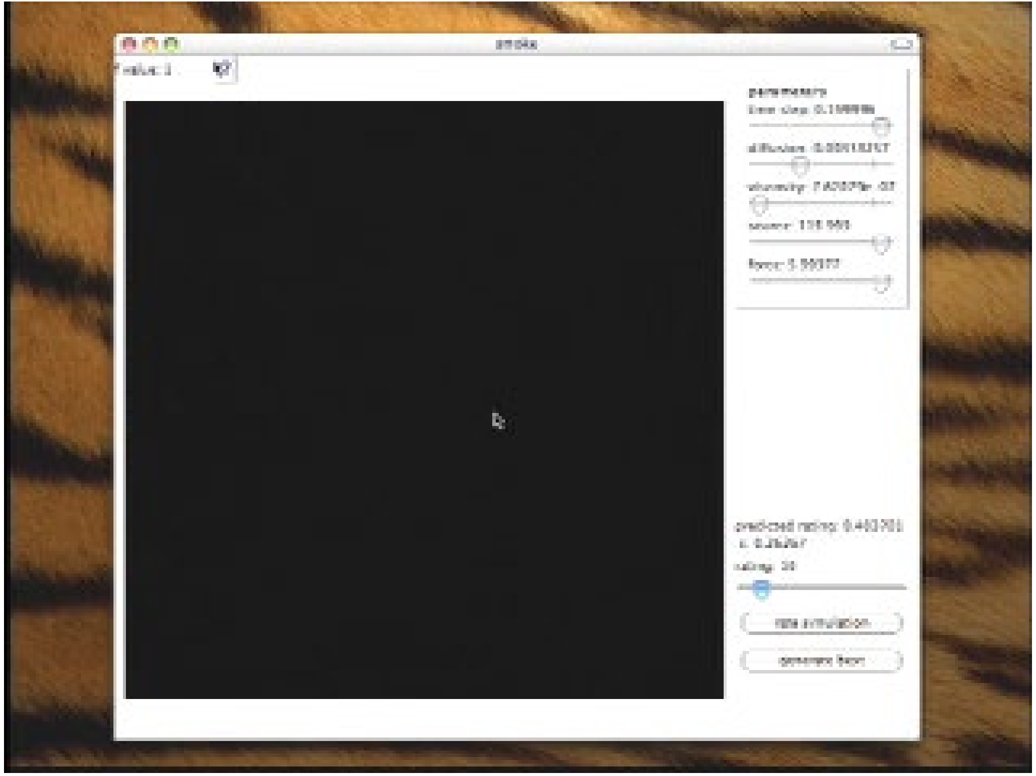
BayesOpt | Original | BayesOpt | Original

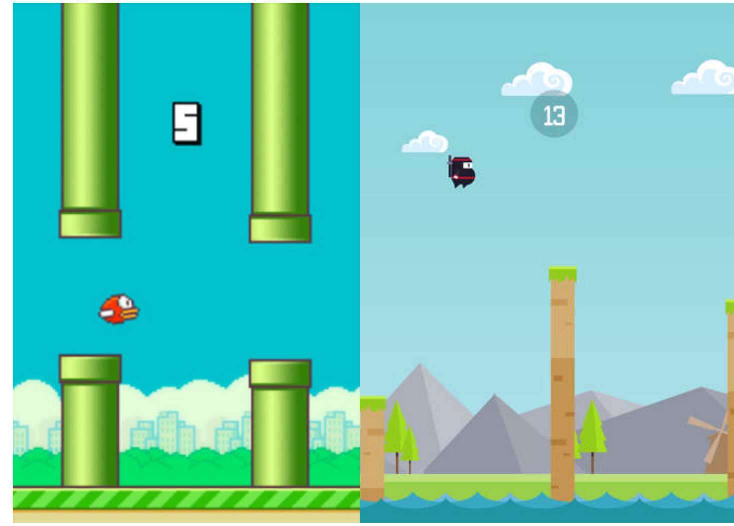


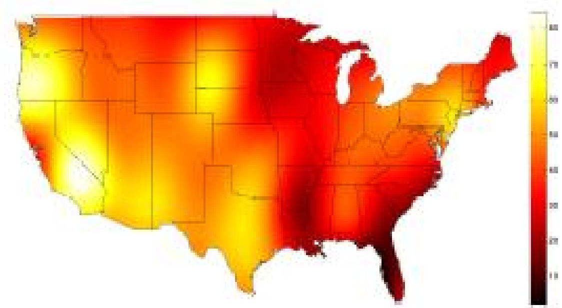
Cell migration



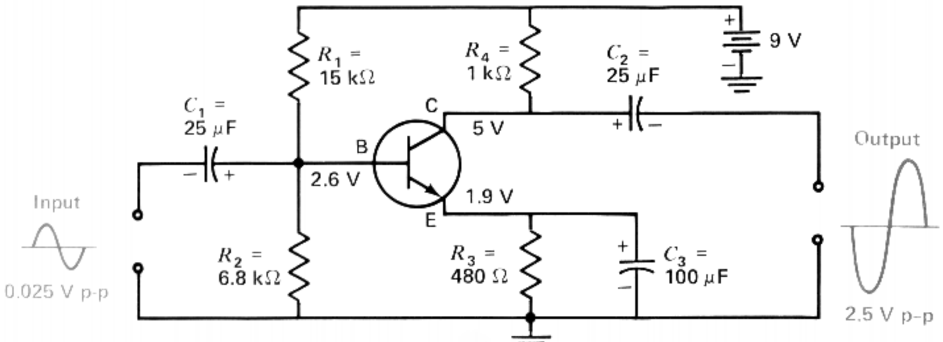




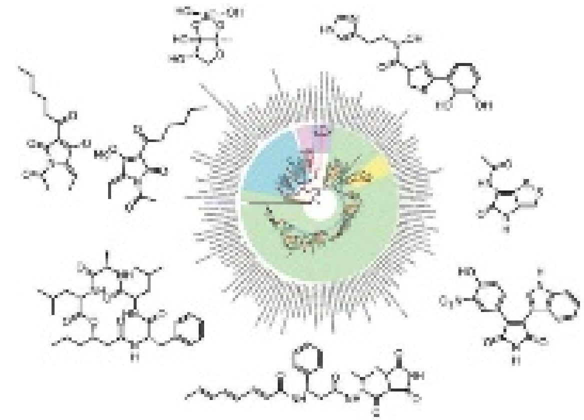


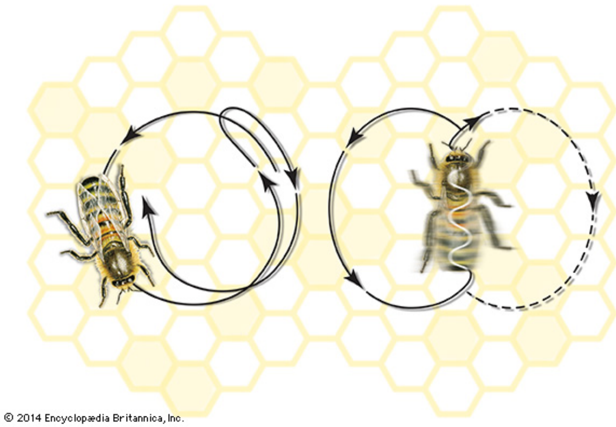


Universidad
Zaragoza
1542

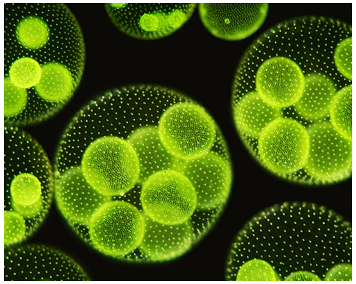


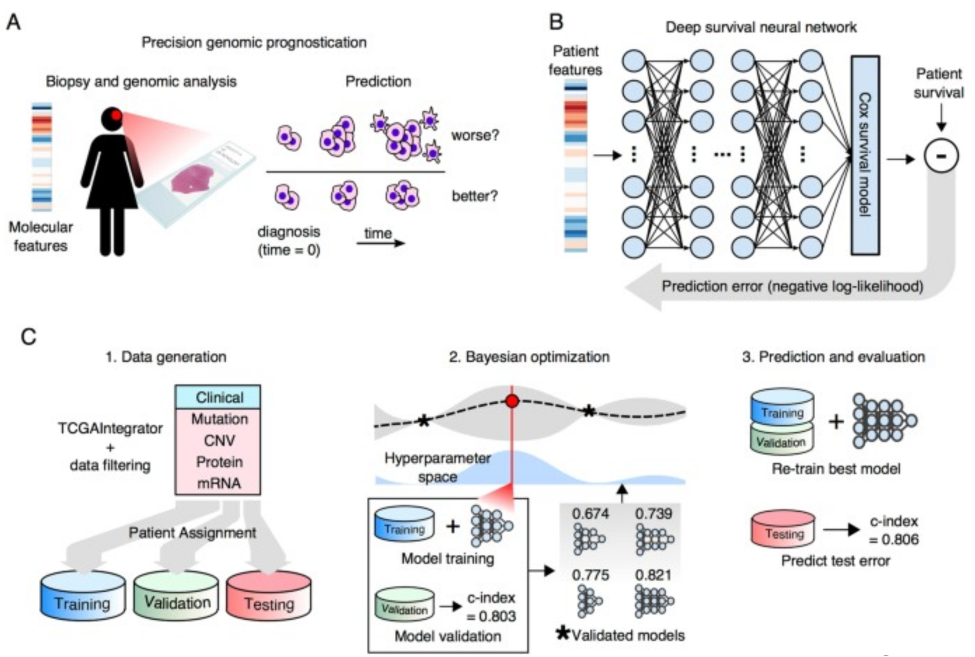
R_1 = 15 kΩ
R_4 = 1 kΩ
C_2 = 25 μF
9 V
C_1 = 25 μF
C
5 V
B
2.6 V
1.9 V
E
Output
Input
0.025 V p-p
R_2 = 6.8 kΩ
R_3 = 480 Ω
C_3 = 100 μF
2.5 V p-p



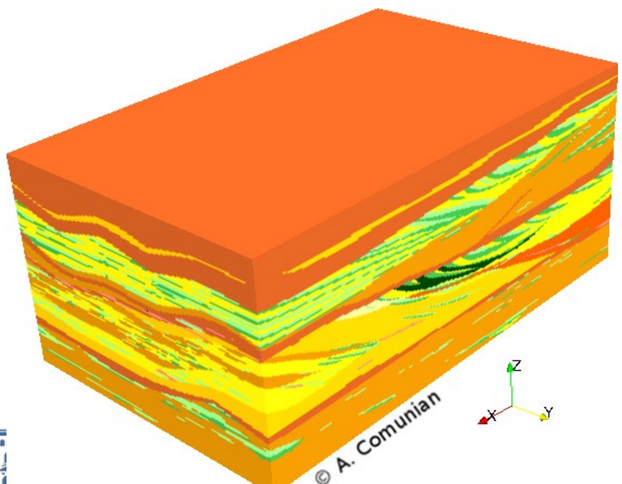


© 2014 Encyclopædia Britannica, Inc.

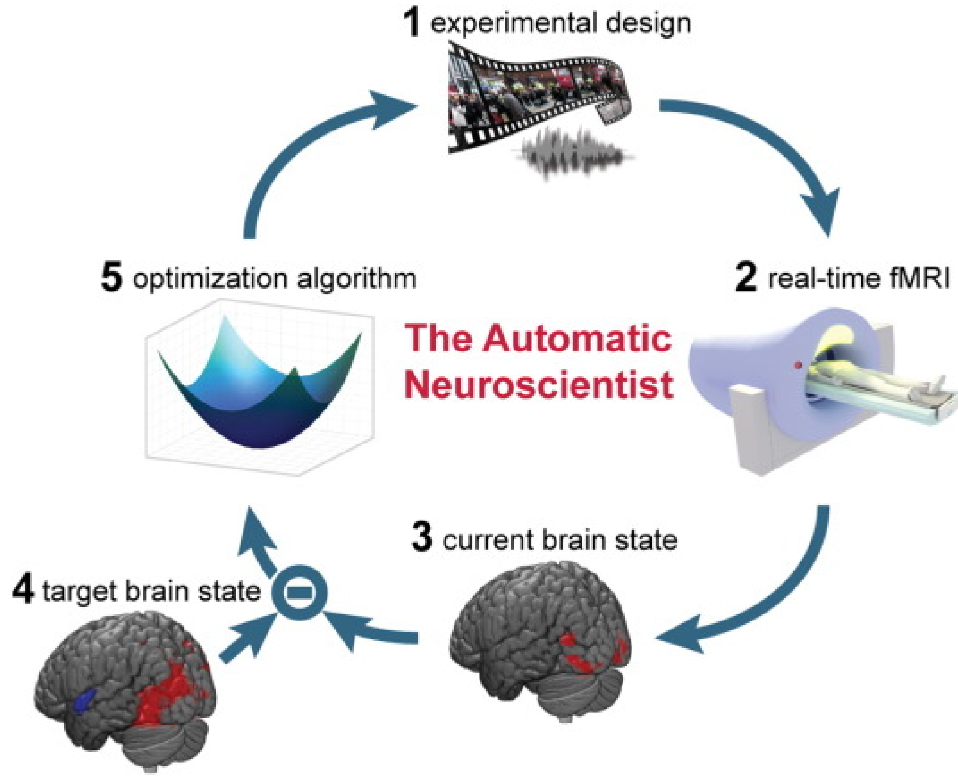




A
Precision genomic prognostication
Biopsy and genomic analysis
Prediction
worse?
better?
Molecular features
diagnosis (time = 0)
time
B
Deep survival neural network
Patient features
Patient survival
Cox survival model
Prediction error (negative log-likelihood)
C
1. Data generation
2. Bayesian optimization
3. Prediction and evaluation
TCGAIntegrator + data filtering
Clinical
Mutation
CNV
Protein
mRNA
Patient Assignment
Training
Validation
Testing
Hyperparameter space
Model training
c-index = 0.803
Model validation
0.674
0.739
0.775
0.821
Validated models
Re-train best model
c-index = 0.806
Predict test error



© A. Comunian
Z
X
Y



1 experimental design
2 real-time fMRI
3 current brain state
4 target brain state
5 optimization algorithm
The Automatic Neuroscientist

A
B
C
D
E
F
G

Universidad Zaragoza 1542

Machine Learning (69152)

Máster in Robotics, Graphics and Computer Vision